

Samarth Agarwal

📍 New York, USA

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PROFESSIONAL SUMMARY

AI Software Engineer with expertise in building production-ready LLM systems, multi-agent architectures, and edge-native AI applications. Proven experience deploying scalable AI solutions using RAG, LoRA fine-tuning, & agentic workflows. Combines strong software engineering fundamentals (Java/Spring Boot, Python, React) with advanced ML deployment techniques to deliver cost-efficient, high-performance AI systems. Open-source contributor, merged security & testing fixes in backend infrastructure, with active contributions across AI/agent tooling.

EXPERIENCE

Generative AI Engineer Intern, ARMA AI Labs

June 2025 - Aug 2025 | California, USA

- Engineered a DeepEval-powered evaluation framework using 7 metrics (BLEU, BERTScore, GEval, AnswerRelevancy, Faithfulness) with custom normalization, resulting in objective prompt comparison across LLMs.
- Revealed GPT-4's **96%** factual accuracy and Gemini's **40x** cost-efficiency by benchmarking models across 20+ articles, resulting in data-driven routing strategies for domain-specific generation.
- Optimized content delivery by delivering a production-ready dynamic routing pipeline, resulting in balanced performance across accuracy, latency, and cost requirements.

Senior Associate Software Developer, Genpact

July 2022 - July 2024

- Reduced manual data entry by **60%** for enterprise clients by engineering scalable Salesforce CRM architectures using Apex triggers and Process Builder, resulting in enforced data integrity.
- Ensured low-latency data consistency globally by designing CORA workflow automation and integrating Salesforce with ERP systems via SFTP and Spring Boot microservices, resulting in synchronized pricing and orders.
- Enhanced component stability by progressing from Intern to Senior Associate and using LWC, SOQL queries, and code reviews, resulting in resolved production defects and improved CRM visualization.

Machine Learning Engineer Intern, TannMann Foundation

June 2021 - Aug 2021

- Achieved **96%** validation accuracy in brand logo detection by optimizing a TensorFlow pipeline using data augmentation and backbone freezing, resulting in mitigated overfitting.

TECHNICAL SKILLS

AI/ML & LLMs: PyTorch, TensorFlow, LangChain, RAG, LoRA Fine-tuning, MCP, DeepEval, Prompt Eng., Agentic Workflows, LLM-as-Judge

Languages & Backend: Python, Java, C++, SQL, Spring Boot, FastAPI, Flask, REST APIs, Hibernate, Apex

Data & Vector DBs: ChromaDB, Pinecone, Pandas, NumPy, Spark, ETL, AWS, Cloudflare Workers, Docker, Git

Frontend & Tools: React, Tailwind CSS, Vite, Pyodide/WASM, Hugging Face, Salesforce CRM, LWC, Visualforce

PROJECTS

ParamMem: Parametric Reflective Memory for LLM Agents [LoRA Fine-tuning, Llama-3, Groq LPU, ChromaDB, HumanEval/GSM8K, HotpotQA]

- Accelerated LLM agent reasoning by replacing retrieval-heavy episodic memory with a LoRA-tuned parametric module on Llama-3, which mitigated reflection collapse and slashed iterative loop latency by **94%** (45.2s to 2.8s) via high-speed Groq API integration.
- Optimized agent memory and reduced token footprint by **45%** by architecting a ParamAgent-plus hybrid system that pairs parametric reflection with ChromaDB vector trajectory retrieval, achieving **79.8%** on the HumanEval coding benchmark and **66.7%** on the GSM8K math dataset.

FlowOps AI: Autonomous Agentic CRM Deal Desk [React, FastAPI, LangGraph, DeepEval, Langfuse, Groq/Gemini, Supabase]

- Engineered an autonomous CRM deal desk by orchestrating a stateful multi-agent LangGraph workflow (Groq/Gemini), which automated BANT qualification while enforcing strict human-in-the-loop (HITL) margin guardrails for enterprise sales.
- Guaranteed production reliability and achieved a **0.93** BANT extraction F1 score by integrating DeepEval for continuous LLM benchmarking across 7 metrics, alongside Langfuse for real-time distributed tracing.

Autonomous Multi-Agent AI Penetration Testing Framework [React, Flask, Groq LPU/Gemini, Multi-Agent, NVD/CVE, Nmap/Shodan, Docker, Cloudflare]

- Automated comprehensive penetration testing, reducing security audit times from days to minutes, by engineering a multi-agent pipeline that orchestrates 4 specialized agents (Recon, Vuln Scanner, Exploit Engine, Strategic Reporter) via Groq LPU Llama-3.1-70B with Gemini 2.5 Flash fallback.
- Delivered professional-grade vulnerability reports with automated CVE mapping via NVD API and CVSS scoring alongside Nmap/Shodan OSINT, while enforcing strict security guardrails with regex target blocking (.gov/.mil), SSRF prevention, and cryptographic audit logging.

Real-Time Financial Data Pipeline: Serverless Edge Evolution [Cloudflare Workers, D1 Edge SQLite, R2, React 18, D3.js, Finnhub, Kafka/Snowflake (V1)]

- Delivered ultra-low-latency financial market data by architecting an end-to-end global serverless pipeline on Cloudflare Workers (V8 isolates), D1 distributed SQL, and R2, successfully transitioning from a legacy enterprise MDS (Kafka, Snowflake, Airflow, Docker).
- Eliminated idle compute costs and reduced external API calls by **90%** by engineering smart on-demand edge caching (< 60s TTL), which delivered sub-60s data latency and global CDN distribution via Cloudflare Pages.

EDUCATION

Stony Brook University SUNY, Master of Science in Data Science

Aug 2024 - May 2026 | New York, USA

Statistical Learning, Data Management, Big Data Algorithms & Networks, Statistical Computing, HCI, Frontiers in LLMs

Amity University, Bachelor of Technology in Computer Science & Engineering

June 2018 - June 2022

DBMS, Operating Systems, Data Structures, OOP, Java, Theory of Computation, Compiler Construction, Big Data, Cyber Security

ACHIEVEMENTS

[Salesforce AI Associate](#) | [Redis Associate Developer](#) | [SAP Generative AI Developer](#) | [AWS Solutions](#) | [BCG Data Science](#) | [GSolve.ai ML Hackathon Winner](#)

| [Geoffrey Hinton Hackathon \(Rank 123\)](#) | [Pneumonia Detection Hackathon \(Top 50 /2k+\)](#) | [Dare in Reality Hackathon\(Rank 159\)](#) | [UHack Hackathon \(Rank 160\)](#)